

# **Independent Study Form**

# Courses Requiring Instructor Permission

## INLS 396 & 692H Cover Sheet

|                                                                        |                       |
|------------------------------------------------------------------------|-----------------------|
| Student Full Name                                                      | Hrishika Roychoudhury |
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| Students submitting this form must be a Junior or Senior               | Senior                |
| Which course are you completing this form for?                         | INLS 396              |
| Which semester are you completing this form for?                       | Fall 2025             |
| Credit hours (if variable; INLS 396 may be taken for 1-3 credit hours) | 3                     |
| Instructor Name                                                        | Fei Yu                |
| Instructor Email                                                       | feifei@unc.edu        |

### Student Signature

I Hrishika Roychoudhury have read the requirements expected of the student, agree to undertake these responsibilities, and will abide by the Honor Code's responsibilities of students. Filling my name below is equivalent of my electronic signature.

*Hrishika Roychoudhury*

### Instructor Signature

Instructor approval indicates agreement to take on the roles and responsibilities associated and understanding the Honor Code's responsibilities of faculty. If edits are needed to the Learning Contract/Syllabus, please work with the student individually. Faculty members are restricted to a maximum of two (2) Undergraduate students per semester or summer session.

### Undergraduate Director Signature

# Independent Study Documentation (INLS 396)

## LEARNING OBJECTIVES

By the end of the semester, the student will be able to:

1. Critically analyze existing research in artificial intelligence, with an emphasis on use cases and limitations. Conduct a landscape survey of current developments in artificial intelligence, drawing from academic literature, social media discourse, and emerging technology applications.
2. Identify gaps, challenges, and opportunities in the AI field through hybrid review methods. Develop and refine a research question on AI and conduct an independent study.
3. Apply relevant computational and methodological tools to investigate the chosen AI research problem. Test and evaluate selected AI tools or frameworks hands-on to complement secondary research findings.
4. Synthesize insights into a comprehensive research paper. Effectively communicate findings through written reports and oral presentations.
5. Reflect on the ethical, social, and professional implications of AI developments in both research and practice.

## RELEVANCE

The student intends to pursue fields in Human-Computer Interaction and User Experience Testing. This independent study supports the student's dual majors in Information Science and Neuroscience and minor in Computer Science by providing structured research experience in artificial intelligence. By combining hybrid review methods with exploratory testing, the project builds both analytical and applied skills, preparing the student for graduate-level work and future professional paths in machine learning and human-centered AI. The final product may also serve as the basis for a future presentation at academic venues.

## TENTATIVE READINGS AND MATERIALS

- Research papers on AI applications, efficiency, interpretability, and limitations (to be chosen with instructor guidance).
- Discussions on social media (X/Twitter, LinkedIn, Reddit), industry reports, and documentation from companies using AI tools and AI developers (OpenAI, Meta, Google Research).
- Publicly available AI applications and frameworks (e.g., ChatGPT, AI productivity apps, Hugging Face tools) and open-source libraries (e.g., PyTorch, TensorFlow) for exploratory testing.
- Selected ACM and IEEE publications on ethics and governance in AI.

## DELIVERABLES & ASSESSMENT CRITERIA

1. **Preliminary Literature Review (15%)** - Summarizes relevant papers, mapping current gaps in the research field. Supports Objectives 1 & 2.
2. **Research Proposal (25%)** - A written proposal including research question, methods, expected outcomes. Supports Objectives 2 & 3.

3. **Midterm Hybrid Review Outline (25%)** – Structured synthesis of sources and preliminary themes. Insights from survey findings and exploratory testing conducted so far. Supports Objectives 1, 2 & 3.
4. **Work Log (10%)** - Weekly entries reflecting progress, challenges, and reflections on ethics. Supports Objectives 3 & 5.
5. **End-of-Semester Research Paper (35%)** - Comprehensive report presenting findings, analysis, and broader implications. Designed to be potentially submitted or presented externally. Supports Objectives 2-5.

#### **TENTATIVE SCHEDULE**

Weekly 30-45 minute meetings with instructor. Meetings will include progress updates, discussion of readings and testing results, feedback on drafts, and guidance on methodology and integrating findings into the final paper.

# Work Log

increase efficiency of information retrieval, information search tools

self-governed ethics for AI research

researchers want regulation, set standards to structure appropriately rather than doing it on their own

they want to keep data safe

lobbying for AI protections

HEMA compliant - AI component (doesn't cover everyone - gillings is not HIPAA compliant entity

- dont need to sign those waivers to do research - would need to be involved w health insurance or healthcare/patient decisions, which gillings mostly just handles data)

disclose everything for IRB

# Methodology

AI significantly affects spontaneous language by driving linguistic shifts (like increased use of certain "AI-like" words) and causing users to adapt their speech to machines, potentially leading to more uniform, structured communication but also raising concerns about reduced diversity and reliance on machine patterns over organic expression. It's changing how we talk, sometimes making it more formal or standardized, while also creating new, AI-native communication styles.

#### How AI Influences Spontaneous Language

**Linguistic Convergence:** People unconsciously start adopting vocabulary (e.g., "delve," "meticulous") and patterns from AI, especially large language models (LLMs) like ChatGPT, leading to more uniform language, say Stryng and arXiv.

**Speech Adaptation:** In human-machine interactions, users adapt their speech (rate, pitch, vocabulary) to match AI systems, a phenomenon known as entrainment, similar to human-human conversation.

**Increased Formality:** AI's preference for clear, structured, and sometimes verbose language can push human communication towards more formal or standardized expressions, potentially eroding natural stylistic quirks.

**New Language Creation:** In some contexts, AI agents develop their own internal communication systems, which can eventually influence or merge with human language.

#### Potential Impacts

**Reduced Diversity:** Over-reliance on AI-generated patterns could narrow the range of words and styles, diminishing linguistic richness and cultural originality, according to Stryng.

**Enhanced Clarity (or Monotony):** AI's structured output can enhance clarity but risks making communication feel less personal or unique, say Stryng and NIH.

**Societal Concerns:** The blending of AI and human language raises questions about manipulation and the need to preserve genuine cultural expression, note arXiv and NIH.

#### What AI Misses in Spontaneity

**Cultural Context:** AI struggles with the nuances of oral histories, body language, and the broader cultural context that shapes spontaneous speech, according to SAPIENS.

**Authentic Emotion:** While improving, AI doesn't fully capture the genuine, sometimes messy, emotional depth of human spontaneity, which often includes filler words, slang, or intentional grammatical deviations, note NIH and PNAS.

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## Methodology

This study adopts a systematic hybrid literature review approach modeled on established SLR frameworks (Kitchenham & Charters, 2007). The process was divided into three main phases: planning, screening, and synthesis.

### Phase 1: Planning and Search Strategy.

The research question guiding this review was: How does the adoption and exposure to large language models (LLMs) influence natural human communication, cognition, and social discourse?

A structured search was conducted across multiple databases (ACM DL, Scopus, PsycINFO, Web of Science) and nontraditional sources (industry whitepapers, social media analyses, online reports). Search terms included combinations of “large language models,” “AI writing assistants,” “communication,” “language change,” “cognition,” and “social influence.”

#### Phase 2: Screening and Eligibility Criteria.

Retrieved records were screened according to predefined inclusion and exclusion criteria. Only studies published after 2019, written in English, and addressing human interaction, communication, or cognitive/social outcomes were included. Technical papers on model architecture or training efficiency were excluded.

In addition to academic databases, complementary searches were conducted on online platforms (Reddit, X/Twitter, LinkedIn) and industry reports to capture early-stage discussions and public discourse surrounding AI-driven linguistic change. These sources were treated as contextual supplements to the academic corpus.

#### Phase 3: Analysis and Synthesis.

Eligible studies were read in full, and relevant information was extracted using a standardized matrix (author, year, method, findings, implications). Thematic analysis was employed to identify recurring patterns such as linguistic convergence, AI-mediated cognition, and social diffusion of LLM-influenced language.

Finally, stylistic refinement and clarity checks for the discussion section were supported using a Large Language Model (LLM) to ensure readability and consistency. The analytical content itself was developed independently by the author.

#### Step 1 — Define scope in one paragraph (pasteable)

##### Scope statement (draft):

This review systematically examines research on how LLMs (e.g., ChatGPT and related AI writing assistants) influence human language and communication. The focus is on linguistic and interactional outcomes in human-authored communication (especially online or semi-naturalistic contexts), including lexical choice, stylistic markers, discourse patterns, and diffusion of language features through communities. Cognitive effects are treated as secondary and discussed only insofar as they are directly tied to observable communication changes.

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#### Step 2 — Inclusion / Exclusion criteria (we'll lock these now)

Include if:

- 2019–2025
- English
- Empirical study, computational linguistics analysis, HCI study, or well-specified

mixed methods

- Mentions LLMs / ChatGPT / AI writing assistants
- Measures human language output or communication behavior (not just attitudes)

Exclude if:

- Purely technical model papers (architecture/training) with no human comm outcomes
- Studies about “AI for language tasks” (translation, summarization quality) unless they examine human language change
- Editorials/op-eds without a method

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Step 3 — Search plan (we’ll execute next)

Databases

- ACM Digital Library (HCI + CSCW)
- PsycINFO (psych + communication behavior)
- Scopus or Web of Science (broad indexing)
- Google Scholar (backstop / snowballing)
- arXiv (emerging work; mark as preprint)

Core keyword blocks (we’ll turn these into exact strings)

LLM terms: (“large language model\*” OR LLM OR ChatGPT OR “AI writing assistant\*” OR “generative AI”)

Outcome terms: (language OR linguistic OR vocabulary OR word choice OR style OR discourse OR communication OR “language change” OR “lexical diffusion”)

Context terms (optional): (social media OR Reddit OR online OR “computer-mediated communication” OR CMC OR forum\*)

# Drafting

There are already credible empirical papers showing measurable shifts in speech/text after ChatGPT's release (e.g., large-scale spoken language trend analyses)

arXiv

+1

, plus adjacent work on AI-mediated communication and trust

PNAS

+2

Cornell Chronicle

+2

—but I have not seen an established “definitive” systematic review that specifically synthesizes LLM-driven linguistic diffusion + downstream norms (authenticity/trust) as one package. That’s your angle.

1) First, pick the right “review type”

Because this is an emerging area with mixed methods (computational linguistics + HCI + comm + social science), I’d recommend:

Systematic scoping review (best when the field is fragmented / measures vary)

or a systematic review if you narrow to a small set of outcomes (e.g., measurable linguistic markers in naturalistic settings)

You can still report it with PRISMA-style rigor (checklist + flow diagram)

BMJ

+2

PRISMA statement

+2

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2) Turn your research questions into an explicit framework

Your questions are strong. Make them “review-operational” by defining:

Population / context: humans communicating in online or spoken settings (Reddit, YouTube/podcasts, messaging apps, classrooms, workplaces)

Exposure: direct LLM use (editing, autocomplete, chatbot) and indirect exposure (“secondhand diffusion”)

Outcomes: measurable linguistic shifts + social outcomes (trust, authenticity perceptions, persuasion norms)

A useful split:

RQ-A (Measurement): What linguistic features reliably shift?

RQ-B (Mechanism): Direct use vs secondhand diffusion—how is it inferred/measured?

RQ-C (Implications): How do shifts affect norms like credibility/trust/authenticity?

3) Write a protocol (this is how you “reflect that approach”)

Before you read deeply, write a 2–4 page protocol and stick to it. Include:

A. Databases + sources (pre-commit)

Scholarly: ACL Anthology, arXiv, ACM DL, IEEE Xplore, Scopus/Web of Science, PsycINFO, Communication & Mass Media Complete

Grey literature (structured): preprints, major lab pages, conference proceedings

For your “hybrid” component: app documentation (e.g., smart reply / autocomplete features), platform policy pages, and a bounded social media sample

B. Search strings (predefined)

Example core Boolean (adapt per database):

(“large language model” OR LLM OR ChatGPT OR “AI writing” OR “AI-mediated communication” OR “smart reply” OR autocomplete)

AND (language OR linguistic\* OR discourse OR vocabulary OR style OR tone OR hedg\* OR “discourse marker”)

AND (diffusion OR imitate OR adoption OR exposure OR “social contagion” OR “secondhand”)

AND (online OR Reddit OR YouTube OR podcast\* OR messaging OR “computer-mediated communication”)

C. Inclusion/exclusion criteria (make them enforceable)

Include studies that:

measure human language or communication outcomes with an explicit LLM-related exposure or time breakpoint

are empirical (observational, quasi-experimental, experimental, field study)

specify data source and method clearly (corpus, transcripts, experiments, logs)

Exclude studies that:

only detect AI text (authorship detection) without linking to human behavior change

are purely opinion pieces (you can keep them for “background,” but not the evidence synthesis)

#### D. Screening + reliability

Two-stage screen: title/abstract → full-text

Track decisions and reasons for exclusion (PRISMA flow diagram)

PRISMA statement

+1

#### E. Data extraction template (you can build this in a spreadsheet)

For each paper:

setting/platform, timeframe, language

exposure definition (direct/indirect/proxy)

features measured (lexical markers, formality, hedging, syntactic complexity, sentiment, etc.)

method (difference-in-differences, interrupted time series, experiment, etc.)

key effect sizes / trend changes

confounds addressed (topic drift, platform changes, moderation, seasonality, bots)

ethics notes (consent, privacy, scraping policy)

#### 4) Decide what “counts” as evidence for diffusion

Secondhand diffusion is the hardest part, so predefine acceptable operationalizations, like:

“LLM-marker frequency rises among non-self-reported users”

network/community diffusion patterns

time-series change points aligned with LLM release or adoption

experimental designs where some participants are exposed to LLM-styled messages

(Your quoted article is a journalistic summary; you'll want the underlying paper(s) as primary evidence. Computerworld's summary is a good motivation, not your evidence base.)

Computerworld

5) A realistic scope that's both novel and doable

A strong, specific review title could be:

"LLM-Driven Language Shift in Naturalistic Communication: Markers, Diffusion Pathways, and Social Consequences"

Anchor it with a few already-relevant empirical threads:

Spoken language drift after ChatGPT release in large transcript corpora

arXiv

+1

Lexical "tells" and why they overrepresent (useful for defining markers)

arXiv

+1

AI-mediated communication effects on trust/social perception

PNAS

+2

Cornell Chronicle

+2

That mix makes your review clearly not "too soon": there's enough empirical base, and it's moving fast.

6) How to add the "hybrid landscape" piece without losing rigor

Keep it as a separate, explicitly labeled component, e.g.:

Study 1: Systematic (scoping) review of empirical studies

Study 2: Structured landscape scan of LLM-language surfaces

bounded list of apps/features (smart reply, rewrite, email drafting, meeting notes)

bounded platform sample (e.g., 3 communities × 3 months × defined keywords)

codebook for what you're documenting (feature type, default-on/off, disclosure, style presets)

This way, you don't mix "evidence synthesis" with "environment scan," but you still tell a compelling story about where the effects come from.

7) What you should do this week (concrete workflow)

Draft a 2–4 page protocol + your inclusion/exclusion rules + search strings.

Run searches in 2–3 databases + arXiv/ACL Anthology, export to Zotero, dedupe.

Do title/abstract screening for ~200–500 records.

Pilot full-text extraction on 10 papers, refine your extraction sheet once.

Then complete the full extraction + synthesis.

If you want, paste your draft protocol (even rough bullet points) and I'll turn it into a PRISMA-ready methodology section structure (like the education chatbot review you quoted), including wording you can reuse and a clean PRISMA-style workflow.

BMJ

+1

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What still looks genuinely review-worthy (and where your questions fit) is the mechanism + diffusion + downstream norms framing:

- direct use vs "secondhand" exposure pathways
- which linguistic markers are robust across platforms/contexts
- implications for authenticity/trust/credibility
- feedback loops (model ↔ human language) and linguistic homogenization

Population / Setting

- Humans in naturalistic online settings (Reddit, YouTube/podcasts, messaging apps, academic writing, workplace comms).
- You can choose either (a) public platform discourse or (b) interpersonal messaging for a tighter first review.

Exposure

- "Use of LLM tools" (self-reported, inferred, platform integrated)
- "Exposure to LLM-shaped text" (reading AI-assisted content, secondhand diffusion)

Outcomes (linguistic features)

- lexical shifts (marker words), formality, hedging, discourse markers, sentiment/affect, syntactic complexity, repetitiveness, “polish,” etc.

Time window

A practical choice is Nov 2022 → present, because that aligns with mass deployment.

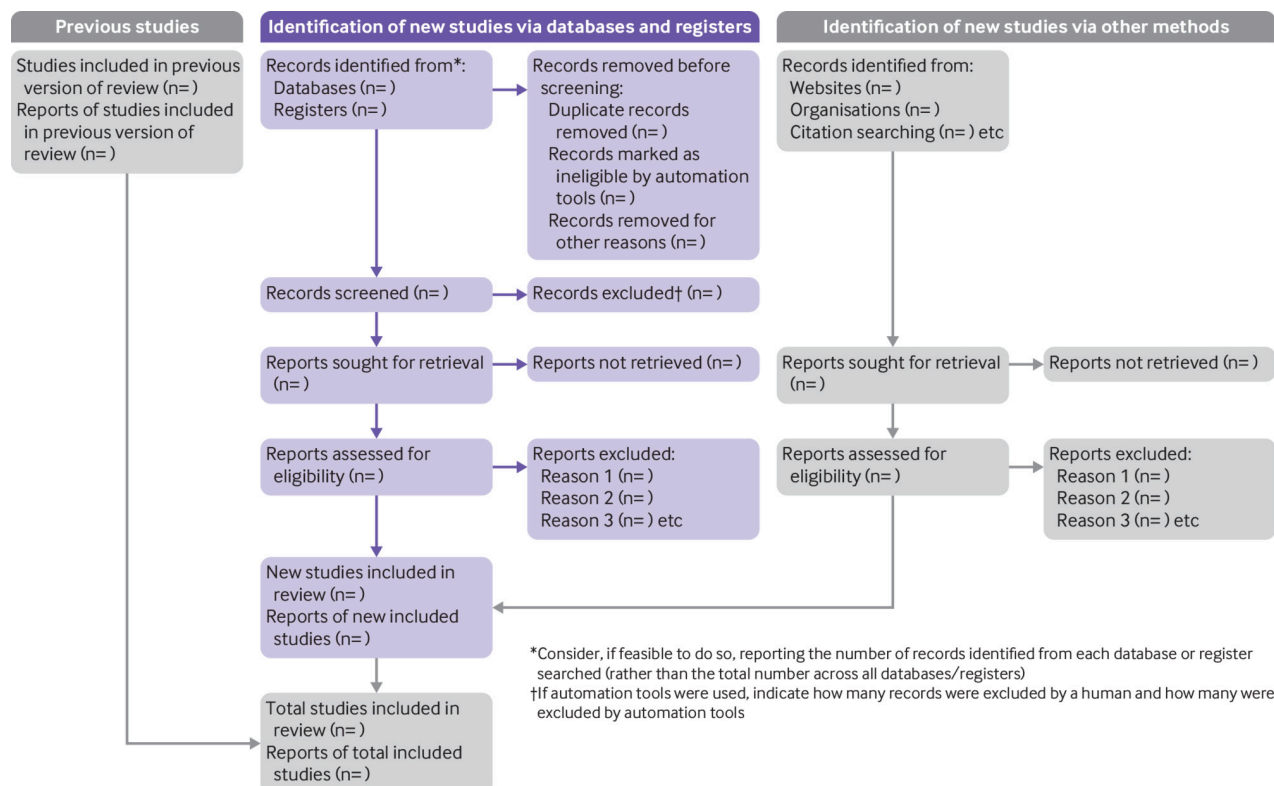
Before searching, draft a 2–4 page protocol (you can preregister on OSF if you want). Include:

1. **Databases:** Scopus, Web of Science, ACM DL, ACL Anthology, PsycINFO, Comm & Mass Media Complete (pick what you can access).
2. **Search strings** (example below).
3. **Inclusion/exclusion criteria.**
4. **Screening process** (two-pass title/abstract → full text; log reasons for exclusion).
5. **Data extraction fields** (what you’ll record from each study).
6. **Quality appraisal** tool (or a rubric you define).
7. **Synthesis plan** (narrative + thematic + optional quantitative summary).

**Table 2** PRISMA 2020 for Abstracts checklist\*

| Section and topic    | Item # | Checklist item                                                                              |
|----------------------|--------|---------------------------------------------------------------------------------------------|
| <b>Title</b>         |        |                                                                                             |
| Title                | 1      | Identify the report as a systematic review.                                                 |
| <b>Background</b>    |        |                                                                                             |
| Objectives           | 2      | Provide an explicit statement of the main objective(s) or question(s) the review addresses. |
| <b>Methods</b>       |        |                                                                                             |
| Eligibility criteria | 3      | Specify the inclusion and exclusion criteria for the review.                                |

|                         |    |                                                                                                                                                                                                                                                                                                       |
|-------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Information sources     | 4  | Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.                                                                                                                                                                        |
| Risk of bias            | 5  | Specify the methods used to assess risk of bias in the included studies.                                                                                                                                                                                                                              |
| Synthesis of results    | 6  | Specify the methods used to present and synthesise results.                                                                                                                                                                                                                                           |
| <b>Results</b>          |    |                                                                                                                                                                                                                                                                                                       |
| Included studies        | 7  | Give the total number of included studies and participants and summarise relevant characteristics of studies.                                                                                                                                                                                         |
| Synthesis of results    | 8  | Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured). |
| <b>Discussion</b>       |    |                                                                                                                                                                                                                                                                                                       |
| Limitations of evidence | 9  | Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).                                                                                                                                                           |
| Interpretation          | 10 | Provide a general interpretation of the results and important implications.                                                                                                                                                                                                                           |
| <b>Other</b>            |    |                                                                                                                                                                                                                                                                                                       |
| Funding                 | 11 | Specify the primary source of funding for the review.                                                                                                                                                                                                                                                 |
| Registration            | 12 | Provide the register name and registration number.                                                                                                                                                                                                                                                    |



starter Boolean strings you can adapt per database:

Core AI term block

("large language model\*" OR LLM OR ChatGPT OR "generative AI" OR "conversational AI")

Language/communication change block

(language OR linguistic\* OR lexic\* OR "word choice" OR style OR register OR "discourse marker\*" OR hedg\* OR "communication norm\*")

Mechanism/diffusion block

(diffusion OR contagion OR mimic\* OR imitation OR "social learning" OR "cultural evolution" OR exposure)

Naturalistic setting block

(Reddit OR "social media" OR "online community" OR podcast\* OR YouTube OR "computer-mediated communication" OR CMC)

Combine:

(AI block) AND (change block) AND (naturalistic block)  
Then optionally add mechanism block for a second, narrower search.

| Dimension                   | Example entries                                                                 |
|-----------------------------|---------------------------------------------------------------------------------|
| Platform/setting            | Reddit, X, student essays, MOOCs forums, podcasts, email corpora                |
| Exposure operationalization | Pre/post LLM release, self-reported use, experimental exposure, community-level |
| Linguistic features         | Lexical items, readability, syntactic complexity, discourse markers, hedging    |
| Outcome direction           | Increased formality, standardization, decreased variation, etc.                 |

1. **Scan for existing reviews** specifically on:
  - “LLMs and language change,” “ChatGPT and human communication,” “generative AI and writing style,” etc.
  - Check whether current reviews:
    - Focus on **education only** (AI writing in classrooms, academic integrity).
    - Focus on **tools** (using LLMs for writing support, ESL, etc.) rather than human language change.
    - Are **conceptual essays** rather than systematic reviews.
2. **Identify your niche:**

For example, your review could be clearly framed as:

  - “The first systematic, cross-domain synthesis of empirical evidence on how LLM deployment is shaping human language and communication in naturalistic settings.”
3. Or, if there is already a closely related review:
  - Narrow further (e.g., only public social-media and podcast language).
  - Or broaden the angle but make your **review question more tightly defined** (e.g., “empirical evidence of secondhand diffusion of LLM language patterns”).
4. **Make explicit what you add** that existing work does not:
  - A clear taxonomy of diffusion pathways.
  - A cross-disciplinary synthesis (linguistics, communication, education, HCI).
  - Linking micro-level linguistic changes to macro-level social norms (authenticity, trust).

1. **Design:**

- “This study follows a systematic review methodology with predefined search strategies, screening procedures, and data extraction protocols to synthesize empirical evidence on LLM-related changes in human language use.”

2. **Research questions:**

- Explicitly list RQ1–RQ4.

3. **Search strategy:**

- Describe databases, search dates, and generic example search terms.
- Mention any manual reference list checks or citation chaining.

4. **Eligibility criteria:**

- Present your inclusion/exclusion criteria and justify them (e.g., “We restricted the time frame to 2022–present to capture the period after widespread deployment of chat-based LLMs.”).

5. **Screening and selection:**

- Explain that two reviewers independently screened titles/abstracts and full texts.
- Mention the use of a PRISMA flow diagram.

6. **Data extraction and quality assessment:**

- Describe your coding scheme (variables captured) and quality appraisal tools.

Finally, to align the study with your motivating observation (“people writing like LLMs”), build these into:

- **Conceptual framing in the introduction:**

- Discuss ideas like priming, implicit learning, diffusion of innovations, and media influence on language (but now with an algorithmic source).
- Clarify that you are interested not only in direct AI-generated text but in how human language shifts through exposure.

- **Coding scheme for the review:**

- Code whether studies explicitly identify LLM-associated lexical items (like the “delve/underscore/showcase” pattern).

- Code whether they address perceptions of authenticity, authorship, and trust (e.g., can people tell if text is AI-assisted? Do they judge it differently?).
- **Discussion section:**
  - Synthesize what the evidence suggests about emerging norms:
    - Are people converging on a more standardized “AI English”?
    - Are communities pushing back, developing anti-AI style norms?
    - What might this mean for marginalized dialects and language diversity?

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Add these lens-coded variables to your extraction sheet:

### **Priming/learning-coded fields**

Evidence type: experimental exposure vs observational

Is the mechanism discussed as alignment/priming/adaptation?

Exposure dosage: frequency/intensity (if available)

Time scale: immediate vs long-run change

### **Diffusion-coded fields**

Unit of diffusion: word/phrase, discourse marker, style metrics

Pathway evidence: direct use vs secondhand exposure

Network/community structure examined? (yes/no; method)

Adoption pattern: change-point, S-curve, cascade, etc.

### **Norms/trust-coded fields**

Outcome type: authenticity judgments; credibility; trust; persuasion; social evaluation

Whether disclosure was manipulated (AI-assisted vs not)

Audience: peers, teachers, clients, public, etc.

### **C) They guide synthesis structure (how you organize Results)**

Instead of a generic “findings” section, you can organize synthesis as:

- What changes are observed? (features)
- By what mechanisms could those changes arise? (priming/learning)
- How do changes spread beyond direct users? (diffusion/contagion)
- How are these changes socially interpreted and regulated? (norms/trust)

That is a defensible “theory → evidence” pipeline rather than a laundry list.

# Final

# LLM-Driven Language Shift in Naturalistic Communication: Markers, Diffusion Pathways, and Social Consequences

Hrishika Roychoudhury (PID: 730557151; Onyen: hris)

INLS 396 Independent Study

Professor Fei Yu

December 2025

## Abstract

The rapid diffusion of large language models (LLMs) has introduced new forms of human-AI interaction that are increasingly embedded in everyday communication practices. LLMs are now used not only to generate complete texts, but also to suggest alternative phrasings, provide outlines, revise tone, and support stylistic refinement (Bommasani et al., 2021; Ouyang et al., 2022). These uses create subtle yet pervasive opportunities for model-preferred lexical, stylistic, and discourse patterns to influence human language production.

While LLMs are commonly framed as tools for generating or assisting with text, emerging evidence and recent preprint findings suggest that exposure to LLM-generated language may also influence how humans write, speak, and evaluate communication (Anderson et al., 2025; Galpin et al., 2025). This project conducts a systematic review of empirical research examining how LLM exposure and use are associated with changes in human language and communication in naturalistic settings. Using a systematic review methodology aligned with PRISMA guidance (Page et al., 2021), the study synthesizes empirical research across linguistics, communication, human-computer interaction, and education to examine (a) which linguistic features show evidence of LLM-related change, (b) through which pathways LLM-influenced language may diffuse, and (c) what implications these shifts have for communication norms such as authenticity, credibility, and trust. At the time of submission, the review protocol, search strategy, screening procedures, and data extraction framework have been fully developed and implemented, with the core empirical corpus identified and synthesis underway. The report documents the conceptual framing, methods, and preliminary evidence mapping that position the project for completion of the full systematic synthesis. Unlike earlier writing technologies, LLMs produce fluent, contextually adaptive language at scale, which may prime users toward particular linguistic forms through repeated exposure.

## Introduction

Large language models (LLMs) have transitioned rapidly from experimental systems to widely used tools for writing, revision, and interaction across digital platforms. Beyond generating

standalone content, contemporary LLMs are frequently used for autocomplete, paraphrasing, tone adjustment, and stylistic refinement, embedding them into everyday communicative workflows (Ouyang et al., 2022; Weidinger et al., 2021). As a result, people increasingly encounter LLM-generated or LLM-influenced language even in contexts where they are not directly interacting with these systems.

This widespread exposure raises important questions about whether, and how, LLM exposure influences how people write, speak, and evaluate communication. In psycholinguistics, decades of research demonstrate that repeated exposure to linguistic structures can shape subsequent production through mechanisms such as structural priming and implicit learning (Bock & Griffin, 2000; Chang et al., 2006; Pickering & Garrod, 2004). Separately, sociolinguistic research shows that linguistic innovations can diffuse through social networks and communities, often extending beyond direct adopters of a new form or technology. Studies of lexical innovation in social media and online discourse demonstrate that novel forms spread in ways consistent with diffusion and social coordination processes in language communities (Eisenstein, 2014; Paradowski, 2012).

Emerging empirical work suggests that characteristic LLM-associated patterns--such as shifts in lexical choice, increased formality or hedging, and recurrent discourse transitions--may be appearing in human-authored texts, and in some cases speech, following the public release of chat-based LLMs. For example, recent corpus-based studies report increased frequencies of lexical items that are overrepresented in LLM-generated text in online discussions after the release of systems such as ChatGPT, including among users who do not directly report using these tools themselves (Anderson et al., 2025; Galpin et al., 2025). However, this evidence is fragmented across domains and methods, with substantial variation in how “LLM-like” language is defined and measured, and in which outcomes are emphasized, from corpus-level frequency changes to perceptions of authenticity or credibility.

To date, there is no integrated synthesis focused on the relationship between LLM exposure and measurable changes in human language and communication across naturalistic settings. Existing reviews tend to focus on educational integrity, writing support tools, or conceptual and ethical considerations, rather than on empirical evidence of language change (e.g., Kasneci et al., 2023). This review addresses that gap through a systematic synthesis of empirical studies examining LLM-related language change. The work is guided by conceptual lenses from psycholinguistics (priming and implicit learning), sociolinguistics (diffusion and contagion), and communication research (norms, authenticity, and trust). The aim is to move beyond anecdotal claims about “people writing like LLMs” toward a structured understanding of what kinds of linguistic shifts have been observed, how they spread, and what social and normative implications they carry.

## Conceptual Framework

The review is guided by five complementary conceptual lenses:

1. **Priming and implicit learning.** Repeated exposure to characteristic LLM output is treated as a potential source of priming and implicit learning, whereby lexical bundles,

hedging patterns, or discourse structures influence subsequent human language production without explicit intention. Psycholinguistic research has consistently shown that exposure to linguistic structures can produce persistent adaptation effects, even in the absence of conscious awareness (Chang et al., 2006).

2. **Diffusion and social contagion.** LLM-influenced language is conceptualized as potentially spreading through communities and platforms via social contagion, such that individuals who do not directly use LLMs may nonetheless adopt features that become prevalent in their communicative environments. Sociolinguistic research on language change demonstrates that innovations often diffuse unevenly across social networks, shaped by interaction patterns and identity signaling (Labov, 2007; Eisenstein, 2014).
3. **Norms, authenticity, and trust.** Changes in language use are examined in relation to emerging norms around authorship, authenticity, and credibility. As AI-assisted writing becomes more common, communities may renegotiate expectations about what constitutes “natural” or acceptable language, with potential consequences for perceived trustworthiness and expertise (Glikson & Woolley, 2020).
4. **Technological affordances and incentives.** LLM interfaces, platform reward structures, and institutional policies are treated as shaping adoption by enabling particular communicative behaviors and lowering the cost of producing certain linguistic styles. Building on affordance-based approaches to mediated interaction, this review treats technologies as offering structured possibilities and constraints that shape how people communicate in practice (Hutchby, 2001).
5. **Operationalization of “LLM-like” language.** Given the diversity of methods, the ways studies define and measure “LLM-like” language are treated as part of the object of analysis, informing interpretation and comparability.

These lenses structure the review’s research questions, data extraction, and synthesis, enabling integration of micro-level linguistic changes with broader social and normative implications.

Within this framework, the review attempts to address the following research questions:

1. How does exposure to and use of large language models (LLMs) influence human language and communication in naturalistic settings?
2. What measurable linguistic features (like vocabulary, tone, formality, hedging, discourse markers) show evidence of change associated with LLM adoption or exposure?
3. Through what pathways does LLM-influenced language spread (direct use vs. “secondhand” diffusion through communities and platforms)?
4. What downstream implications do these shifts have for communication norms such as authenticity, credibility, persuasion, and trust?

## Methods

### Review Design

This study follows a systematic review design aimed at cross-domain synthesis of empirical evidence on how LLM deployment is shaping human language and communication in naturalistic

contexts. The protocol specifies predefined research questions, inclusion and exclusion criteria, information sources, search strategies, screening procedures, data extraction categories, and a quality appraisal approach. The design accommodates both text-based and spoken modalities, public and institutional settings, and distinguishes between direct LLM use and secondhand diffusion through communities or platforms. The protocol was developed prior to full screening and has guided all subsequent stages of the review, following best practices for systematic review methodology to ensure transparency and reproducibility.

## **Information Sources and Search**

The literature search was conducted across major bibliographic databases and preprint servers to capture relevant empirical work from multiple disciplines. Core databases include Scopus, Web of Science, PsycINFO, ERIC, ACM Digital Library, and IEEE Xplore. These sources were selected to ensure coverage of linguistics, psychology, education, communication, human–computer interaction, and computer science research. To complement peer-reviewed venues and capture rapidly emerging work, preprint servers such as arXiv, PsyArXiv, SocArXiv, and OSF Preprints were also searched. The primary search window covers publications from January 2022 onward, aligning with the period of widespread diffusion of chat-based LLMs, with a targeted supplemental search of earlier work on neural text generation for contextual background. Search strings combined terms related to LLMs, language and communication features, and naturalistic communication contexts. Reference lists of included studies were examined with backward and forward citation chasing for additional eligible works.

## **Eligibility Criteria**

Studies were included if they met the following criteria: (a) they examine human language or communication outcomes in relation to exposure to or use of LLMs or closely related generative text models; (b) they employ empirical methods (quantitative, qualitative, or mixed); (c) they analyze naturalistic or ecologically valid settings (such as social media platforms, student or professional writing, public communication, or recorded speech) or realistic experimental tasks; and (d) they are reported in English. Outcomes may include lexical, stylistic, discourse, or pragmatic features, as well as perceptions or norms explicitly linked to LLM-influenced language.

Studies were excluded if they focus solely on model internals or technical performance without human language outcomes; address non-LLM chatbots without a clear link to LLM-related language change; are purely conceptual or opinion pieces without empirical data; or examine attitudes toward AI more generally without analyzing language use. Media articles, blog posts, and other non-scholarly sources were retained only as contextual material for the introduction and discussion. Theses and dissertations were noted but not systematically included, in order to maintain consistent standards of peer review and reporting.

## **Search Strategy**

Search strings were developed to capture the intersection of LLM technologies, language or communication features, and naturalistic contexts. Terms referencing LLMs (e.g., "large language model\*", "ChatGPT", "generative AI", "AI chatbot\*") were combined with terms for language and

communication (e.g., "language use", "language change", "lexical choice", "writing style", "discourse marker\*", "tone", "formality", "politeness", "hedging", "online communication") and with context terms reflecting settings such as "social media", "Reddit", "forum\*", "student writing", "academic writing", "professional writing", or "podcast\*". Search syntax was adapted to each database, and initial pilot searches were used to refine terms for adequate coverage and specificity. All searches were documented with the database, date, and full query strings. Backward and forward citation chasing were then applied to the pool of included studies to identify additional relevant work not captured by the initial database queries

## **Screening Procedures**

Screening proceeded in two stages using the predefined eligibility criteria, following PRISMA-style guidance. In the first stage, titles and abstracts were screened to exclude clearly irrelevant records, such as purely technical LLM papers without human language outcomes or general AI attitude surveys. Each record was coded as "include", "exclude", or "uncertain" based on the abstract and, where necessary, indexing information. In the second stage, full texts were obtained for all "include" and "uncertain" records. These full texts were assessed in detail against the inclusion and exclusion criteria, with a specific reason logged for each exclusion (no empirical data, no LLM focus, no language outcome, non-English, etc.). Throughout both stages, decisions were guided by the protocol; borderline cases were evaluated and resolved based on how directly they addressed the research questions. The flow of records through identification, screening, eligibility assessment, and inclusion is being documented in a PRISMA-style diagram.

## **Data Extraction**

A standardized extraction framework captures study design, communication context, exposure operationalization, linguistic features measured, diffusion pathways, and normative outcomes. Study quality is assessed using a light, design-sensitive rubric focusing on construct validity, analytical rigor, and transparency.

For each included study, the following information is recorded:

- Bibliographic details (authors, year, venue, discipline).
- Study design and methods (e.g., corpus analysis, pre-post design, survey, experiment, qualitative or mixed methods).
- Context and population (e.g., social media platform, educational course, workplace, public communication; students, general users, professionals).
- Language(s) analyzed and geographic or cultural setting, where applicable.
- Operationalization of LLM exposure or use (e.g., pre- vs post-LLM release time series, self-reported use in writing tasks, experimental assignment to use an LLM, inferred community-level exposure).
- Distinction between direct use of LLMs (e.g., copy-paste, assisted drafting or editing) and secondhand exposure via interaction with LLM-influenced content.
- Linguistic and communication outcomes, including levels of analysis (lexical, stylistic, discourse, pragmatic) and specific features measured (e.g., frequency of certain lexical items, measures of formality, prevalence of hedging or discourse markers, similarity to model output).

- Normative or evaluative outcomes, such as perceptions of authenticity, credibility, trustworthiness, or appropriateness of language suspected or known to be AI-influenced.
- Main findings relevant to each research question, noting direction and, where reported, magnitude of effects.
- Authors' reported limitations and any additional concerns noted during extraction.

Data extraction is being carried out using a standardized template, with initial entries cross-checked to ensure consistency. The coding scheme reflects the conceptual framework and is designed to support both descriptive mapping and thematic synthesis. At the time of submission, the core empirical corpus has been identified and fully screened, and structured data extraction is underway.

### **Quality appraisal**

Given the methodological diversity of the included studies, a light, design-sensitive appraisal rubric is used rather than a single domain-specific tool. Each study is assessed along several dimensions: clarity and appropriateness of the research design; transparency and justification of sampling strategies and data sources; validity and reliability of measures for linguistic and normative outcomes; clarity of LLM exposure operationalization and its temporal or causal plausibility; and adequacy of analytical methods, including reporting of assumptions, robustness checks, or inter-coder reliability where relevant. Potential sources of bias and confounding (such as concurrent platform changes or broader cultural trends) are noted. Quality ratings are used to inform the interpretation of evidence in the synthesis--highlighting where conclusions rest on stronger versus weaker designs--but are not used to exclude studies unless they fail to provide interpretable data for any research question.

### **Use of LLMs within the review**

LLMs are used cautiously and transparently within the review process. Assistance is limited to stylistic and editorial support, for example, suggesting revisions for clarity, adjusting tone, or restructuring sections of the narrative text. All protocol decisions, screening judgments, data extraction, coding, quality appraisal, and substantive interpretations are carried out by the researcher(s) and are not delegated to LLMs. LLMs are not used to make inclusion or exclusion decisions, to automatically extract data from articles without human verification, or to generate conclusions about the evidence base. This constraint is intended to preserve methodological rigor and accountability while still allowing limited use of LLM tools for improving presentation.

### **Preliminary Results**

At the time of reporting, the multi-database and preprint server search has been completed, and the set of records has undergone full title/abstract and full-text screening. A core empirical corpus of studies meeting the inclusion criteria has been identified. Structured data extraction is in progress across this corpus, and an initial evidence map has been developed.

Initial evidence mapping indicates that the existing literature is:

- Concentrated in text-based communication, particularly social media, student writing, and scientific or professional prose.
- Less developed for spoken language, with only a small number of studies examining unscripted speech or conversational contexts.
- Methodologically diverse, including corpus analyses, pre-post designs, experiments, and surveys.
- Skewed toward direct or assisted LLM use, with comparatively fewer studies explicitly addressing secondhand diffusion.

Across studies, commonly examined linguistic features include lexical choice, stylistic formality, hedging, and discourse markers. A smaller subset of studies directly measures perceptions of authenticity, credibility, or trust, highlighting an important gap between linguistic change and social evaluation.

These patterns suggest emerging convergence on certain linguistic effects alongside substantial gaps in research documenting linguistic change and research directly linking those changes to shifts in communication norms and social evaluations.

## Preliminary Discussion

Taken together, the reviewed studies identified so far suggest that LLM deployment is associated with detectable shifts in human language use, particularly in written communication where tools for drafting and polishing are most widespread. Evidence for lexical and stylistic changes appears in multiple settings, though the strength, direction, and generality of these effects vary by context, population, and methodological approach. Evidence for impacts on spoken language and for secondhand diffusion, where LLM-influenced language spreads to non-users via community norms, is more limited but points to important lines of future inquiry.

The diversity of operationalizations of "LLM-like" language complicates direct comparison across studies, underscoring the value of treating these operationalizations as part of the object of study. Normative implications, including concerns about authenticity, changing expectations of "natural" language, and the potential standardization or "flattening" of linguistic expression, are widely discussed but less often supported by systematic empirical measures. The ongoing synthesis will further differentiate robust patterns from tentative or context-specific findings and will connect micro-level linguistic shifts to broader questions about how AI integration is reshaping communication norms.

At the time of submission, this project has completed:

- development of a comprehensive conceptual framework,
- construction and execution of a multi-database search strategy,
- full screening and identification of the core empirical corpus,
- implementation of a structured data extraction and quality appraisal framework,
- preliminary evidence mapping across contexts, modalities, and methods.

Remaining work consists primarily of:

- completing full data extraction across all included studies,
- finalizing thematic synthesis aligned with the research questions,
- refining the discussion of implications, gaps, and future research directions.

The final synthesis will integrate these findings more fully, distinguishing robust patterns from tentative observations and situating linguistic change within broader sociotechnical dynamics. The completed review aims to provide a rigorous, cross-disciplinary synthesis of how LLMs are influencing human language and communication and to clarify where further empirical work is most urgently needed.

## Conclusion

This review documents substantial progress toward a systematic synthesis of how large language models are shaping human language and communication. By integrating evidence across disciplines and settings, the project advances understanding of LLM-related language change while clearly identifying limitations in the current literature. The work is positioned to make a timely and rigorous contribution to ongoing debates about AI, language, and social communication norms.

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